

Operation

Flush the Pump Before First Use

The pump is tested with lightweight oil, which is left in to protect the pump parts. If the fluid you are using may be contaminated by the oil, flush it out with a compatible solvent. See **Flushing** on page 10.

Starting and Adjusting the Pump

WARNING



MOVING PARTS HAZARD

See Fig. 3. The priming piston (23) and the air motor piston (located behind the air motor plates) move during operation.

Keep hands and fingers away from the priming piston (23) during operation and whenever the pump is charged with air. The priming piston extends beyond the intake cylinder (20) to pull material into the pump and can amputate a hand or finger caught between it and the intake cylinder. Follow the **Pressure Relief Procedure** on page 8, before checking, clearing, or cleaning the priming piston to prevent the pump from starting accidentally.

Never operate the pump with the air motor plates removed.

See the **Installation** on page 7. If you are using a pneumatic elevator cart or 19 liter (5 gallon) ram with the pump, refer to the separate instructions for those components for set-up and operation instructions.

Lower the pump into a fluid container. Be sure the air regulator (K) and bleed-type master air valve (J) are closed. Then open the bleed valve (J). Hold a metal part of the spray gun/dispensing valve (S) firmly to the side of a grounded metal pail and hold the trigger open. Now slowly open the air regulator until the pump starts.

Cycle the pump slowly until all the air is pushed out and the pump and hoses are fully primed. Release the spray gun/dispensing valve trigger and engage the safety latch. The pump should stall against pressure when the trigger is released.

If the pump fails to prime properly, open the bleeder valve (33) slightly. See Fig. 7 on page 16. Use the bleeder hole (on the underside of the valve housing) as a priming valve until the fluid appears at the hole. Close the bleeder valve.

NOTE: When changing fluid containers with the hose and gun already primed, be sure to open the bleeder valve (33), to assist in priming the pump and venting air before it enters the hose. Close the valve when all air has been eliminated.

WARNING

SKIN INJECTION HAZARD

To reduce the risk of fluid injection, **do not** use your hand or fingers to cover the bleed hole on the underside of the bleeder valve body when priming the pump. Use a crescent wrench to open and close the bleeder plug. Keep your hands away from the bleed hole.

With the pump and lines primed, and with adequate air pressure and volume supplied, the pump will start and stop as the spray gun/dispensing valve is opened and closed.

Use the air regulator (K) to control the pump speed and the fluid pressure. Always use the lowest air pressure necessary to get the desired results. Higher pressures cause premature tip/nozzle and pump wear.

WARNING

To reduce the risk of overpressurizing the system, which could result in component rupture and cause serious injury, **NEVER** exceed the specified maximum incoming air pressure to the pump (see the **Technical Data** on pages 24–27).

CAUTION

Do not allow the pump to run dry. It will quickly accelerate to a high speed, causing damage. A pump runaway valve (G), which shuts off the air supply to the pump if the pump accelerates beyond the pre-set speed, is available. See the Typical Installation in Fig. 2. If your pump accelerates quickly, or is running too fast, stop it immediately and check the fluid supply. If the supply container is empty and air has been pumped into the lines, refill the container and prime the pump and the lines with fluid, or flush and leave it filled with a compatible solvent. Be sure to eliminate all air from the fluid system.

Operation

Shutdown and Care of the Pump

For overnight shutdown, stop the pump at the bottom of the stroke to prevent fluid from drying on the exposed displacement rod and damaging the throat packings. Relieve the pressure.

Always flush the pump before the fluid dries on the displacement rod. Refer to **Flushing** below.

Flushing

**WARNING**

**FIRE AND EXPLOSION HAZARD**
Before operating the pump, read the section **FIRE AND EXPLOSION HAZARD** on page 4. Be sure the entire system and flushing pails are properly grounded. Refer to **Grounding** on page 6.

Flush with a fluid that is compatible with the fluid you are pumping and with the wetted parts in your system. Check with your fluid manufacturer or supplier for recommended flushing fluids and flushing frequency. Always flush the pump before fluid dries on the displacement rod.

WARNING

To reduce the risk of serious injury whenever you are instructed to relieve pressure, always follow the **Pressure Relief Procedure** on page 8.

1. Relieve the pressure.
2. Remove the spray tip/nozzle from the gun/valve.
3. Hold a metal part of the gun/valve firmly to the side of a grounded *metal* pail.
4. Start the pump. Always use the lowest possible fluid pressure when flushing.
5. Trigger the gun/valve.
6. Flush the system until clear solvent flows from the gun/valve.
7. Relieve the pressure.

Troubleshooting

WARNING

To reduce the risk of serious injury whenever you are instructed to relieve pressure, always follow the **Pressure Relief Procedure** on page 8.

1. Relieve the pressure.
2. Check all possible problems and causes before disassembling the pump.

PROBLEM	CAUSE	SOLUTION
Pump fails to operate.	Restricted line or inadequate air supply	Clear*; see Technical Data on pages 24–27.
	Obstructed fluid hose or gun/valve; fluid hose ID is too small.	Clear*; use a hose with a larger ID.
	Fluid dried on the displacement rod.	Clean; see Service on pages 12–19.
	Dirty or worn air motor parts.	Clean or repair; see separate air motor manual.
Pump operates, but output is low on both strokes.	Restricted line or inadequate air supply.	Clear*; see Technical Data on pages 24–27.
	Obstructed fluid hose or gun/valve; fluid hose ID is too small.	Clear*; use a hose with a larger ID.
	Bleeder valve is open.	Close.
	Air is leaking into the supply container.	Check wiper plate seal.
	Fluid is too heavy for pump priming.	Use bleeder valve (see page 9); use wiper plate with ram or pneumatic elevator cart.
	Worn packings in displacement pump.	Replace packings.
Pump operates, but output is low on down-stroke.	Fluid too heavy for pump priming.	Use bleeder valve (see page 9); use wiper plate with ram or pneumatic elevator cart.
	Held open or worn intake valve or seals.	Clear valve; replace seals.
Pump operates, but output is low on up-stroke.	Held open or worn piston valve or seals.	Clear valve; replace seals.
Erratic or accelerated pump speed.	Exhausted fluid supply.	Refill and prime.
	Fluid is too heavy for pump priming.	Use the bleeder valve (see page 9); use wiper plate with ram or pneumatic elevator cart.
	Held open or worn piston valve or seals.	Clear valve; replace seals.
	Held open or worn priming piston.	Clear; service.
	Worn packings in displacement pump.	Replace packings.

* To determine if the fluid hose or gun is obstructed, follow the **Pressure Relief Procedure** on page 8. Disconnect the fluid hose and place a container at the pump fluid outlet to catch any fluid. Turn on the air just enough to start the pump (about 20–40 psi [0.14–0.3 MPa, 1.4–2.8 bar]). If the pump starts when the air is turned on, the obstruction is in the fluid hose or gun.

NOTE: If you experience air motor icing, contact your Graco distributor for assistance.

Service

Required Tools

- Torque wrench
- Bench vise, with soft jaws
- Rubber mallet
- Hammer
- Razor or other sharp tool
- 8 mm (5/16 in.) dia. brass rod
- 8 mm, 11 mm, 15 mm, 25 mm, 3/4 in., and 7/8 in. open-end wrenches
- 17 mm box or socket wrench
- 22 mm crow's-foot wrench
- Adjustable wrenches
- 24 in. adjustable wrench
- Thread lubricant
- Anaerobic thread sealant

Disconnecting the Displacement Pump

1. Flush the pump, if possible. Stop the pump at the bottom of its stroke.

WARNING

To reduce the risk of serious injury whenever you are instructed to relieve pressure, always follow the **Pressure Relief Procedure** on page 8.

Keep hands and fingers away from the priming piston during operation and whenever the pump is charged with air. On the pump downstroke the priming piston extends beyond the intake cylinder to pull the material into the pump. The priming piston works under extreme force. During operation and whenever the pump is charged with air, the priming piston can severely injure or amputate a hand or finger, or break a tool, caught between it and the intake cylinder.

The air motor piston (located behind the air motor plates) also moves when air is supplied to the motor. NEVER operate the pump with the air motor plates removed.

2. Relieve the pressure.
3. Disconnect the air hose. Hold the fluid outlet fitting (7) with an adjustable wrench to keep it from being loosened while you disconnect the fluid hose. Remove the pump from its mounting. Note the relative position of the pump's fluid outlet to the air inlet.
4. Using an adjustable wrench, unscrew the coupling nut (104) from the connecting rod (103). Remove the coupling collars (105). Take care not to lose or drop them. See Fig. 4.
5. Hold the tie rod flats with an 11 mm wrench to keep them from turning. Use the wrench (113) provided with the pump (or a 17 mm box or socket wrench) to unscrew the locknuts (106) from the tie rods (102). Carefully pull the displacement pump (107) off the air motor (101).
6. Refer to page 14 for displacement pump service. To service the air motor, refer to the separate air motor manual, supplied.

Service

Reconnecting the Displacement Pump

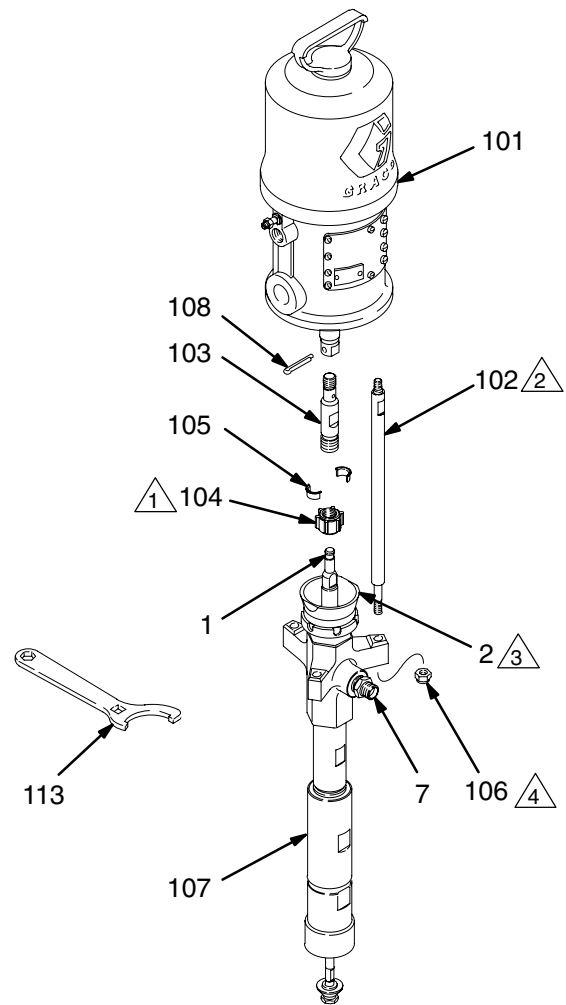
WARNING

To reduce the risk of serious injury whenever you are instructed to relieve pressure, always follow the **Pressure Relief Procedure** on page 8.

1. Orient the pump's fluid outlet to the air motor's air inlet as was noted in step 3 under **Disconnecting the Displacement Pump**. Position the displacement pump (107) on the tie rods (102). See Fig. 4.

NOTE: If you removed the tie rods (102) from the air motor (101) or air motor adapter plate, reinstall them using an 11 mm wrench. Torque the rods to 20-25 N.m (15-18 ft-lb) on Model 222907, and 34-41 N.m (25-30 ft-lb) on Model 222839.

2. Screw the locknuts (106) onto the tie rods (102) and torque to 34-41 N.m (25-30 ft-lb), using the wrench (113) provided (or a 17 mm box or socket wrench).
3. Place the coupling nut (104) on the displacement rod (1), then place the collars (105) in the nut. Screw the nut onto the connecting rod (103) loosely. Hold the connecting rod flats with an adjustable wrench to keep it from turning. Using an adjustable wrench, torque the nut to 41-48 N.m (30-35 ft-lb).
4. Using a torque wrench in the square hole of the supplied wrench (113), torque the packing nut (2) to 27-34 N.m (20-25 ft-lb).
5. Mount the pump and reconnect all hoses. Reconnect the ground wire if it was disconnected during repair. Turn on the air and run the pump slowly.
6. Fill the packing nut/wet-cup 1/3 full of Graco Throat Seal Liquid or compatible solvent.



- 1 Torque to 30-35 ft-lb (41-48 N•m).
- 2 Torque to 15-18 ft-lb (20-25 N•m) on model 222907; 25-30 ft-lb (34-41 N•m) on model 222839.
- 3 Torque to 20-25 ft-lb (27-34 N•m).
- 4 Torque to 25-30 ft-lb (34-41 N•m).

01059

Fig. 4

Service

Disassembly

When disassembling the pump, lay out all removed parts in sequence, to ease reassembly. Refer to Fig. 7.

NOTE: Repair Kit 222784 is available separately to replace the piston and intake valve seals. For the best results, use all the new parts in the kit. Kit parts are marked with one asterisk, for example (8*). See **Displacement Pump Parts** on page 23.

Repair Kit 222785 is available to replace the throat packings. For the best results, use all the new parts in the kit. Kit parts are marked with a check mark, for example (3✓).

1. Disconnect the displacement pump from the air motor as explained on page 12.
2. Place the pump in a vise, with the jaws on the outlet housing (9).
3. Hold the flats of the priming piston rod (21) with an 8 mm wrench. Using a 3/4 in. wrench, unscrew the priming piston nut (22) from the rod. Slide the priming piston (23) and priming piston guide (24) off the rod. Inspect the surfaces of the guide (24) and piston (23) for scoring, wear, or other damage.
4. Loosen the packing nut (2) using the packing nut wrench (113, supplied), or a hammer and brass rod.
5. Remove the intake cylinder (20), using a 24 in. adjustable wrench. Remove the seal (19) and seat (18) from the intake valve housing (16); always replace the seal with a new one.
6. Unscrew the intake valve housing (16) from the cylinder (11), using a 24 in. adjustable wrench. Pull the housing off the pump. The intake check valve assembly (U) should slide down the priming piston rod (21) as you remove the housing; if it does not slide easily, loosen it by firmly tapping the top of the housing (16) with a rubber mallet. Take care not to drop the assembly (U) as it comes free. Set the check valve assembly aside for disassembly later.
7. Inspect the rod guide (17) while in place in the housing (16). Remove the rod guide **only** if it is visibly damaged. Using a hammer and brass rod, gently tap around the **outer edge** of the guide to loosen it.
8. Push the displacement rod (1) down as far as possible, then pull it and the priming piston rod (21) out of the outlet housing (9) and cylinder (11).
9. Remove the packing nut (2) and throat packings from the outlet housing (9). **Do not** remove the fluid outlet nipple (7) and o-ring (8) unless damaged.
10. Unscrew the plug (AA) of the bleeder valve (33) completely from the valve housing (BB). Do not unscrew the valve housing from the outlet housing (9), because the threads may gall. Clean the valve threads and the bleed hole on the underside of the valve housing.
11. Use a 24 in. adjustable wrench on the pump cylinder (11) flats to unscrew it from the housing (9). Remove the o-rings (10). Inspect the inside of the cylinder for damage by holding it up to the light at an angle or running a finger over the surface.
12. Inspect the surface of the displacement rod (1) and priming piston rod (21) for wear, scoring or other damage by running a finger over the surface.
13. Use a vise with soft jaws to hold the displacement rod (1) by its flats. Place a 15 mm wrench on the piston flats and unscrew the piston (13) and priming piston rod (21) from the displacement rod (1). Remove the piston guide assembly (V) from the piston (13), and set aside for disassembly later.
14. It is not necessary to remove the priming piston rod (21) from the piston (13) unless you see scoring, wear, or other damage to either part. To disassemble, place the piston flats in a vise. Using an 8 mm wrench on the rod flats, unscrew the rod.